Change Risk Forecasting & Outage Prediction using Agentic AI

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Theme

Innovation in SRE & Observability using Agentic AI

In today's fast-moving engineering environments, maintaining reliability while innovating rapidly is a constant challenge. Our hackathon entry embraces the theme of "Innovation in SRE & Observability using Agentic AI" by leveraging autonomous AI agents to bring a new level of proactivity and transparency to change management.

Hackathon Topic

Change Risk Forecasting & Outage Prediction using Agentic AI

We propose an AI-driven workflow that analyzes every incoming change request—be it code, infrastructure, or configuration—and predicts its risk of causing service degradation or outages. Beyond a simple score, our system offers clear reasoning, actionable recommendations, and an explainability dashboard to guide teams through safe deployments.

Industry Challenge

High-velocity engineering teams deploy changes constantly, yet each deployment carries the possibility of introducing performance regressions, misconfigurations, or full-blown outages. Traditional rule-based risk assessments are brittle: they miss contextual nuances, fail to learn from historical incidents, and lack the explainability teams need to trust automated recommendations. The result is either overly conservative processes that slow innovation or blind spots that lead to costly downtime.

Proposed Solution Approach

How can we build an autonomous AI agent that analyzes a change request and forecasts its risk level intelligently—complete with reasoning, recommendations, and explainability—before the change is deployed?

1. Intelligent Change Ingestion

- **Natural-Language Parsing:** Agent ingests the change description, justification, and planned timing.
- **Contextual Enrichment:** Ties the request to relevant configuration items (CIs) and past incidents via a knowledge graph.
- **Initial Risk Estimate:** Generates a "low/medium/high" risk label based on criteria such as business-criticality and deployment window.

2. **Historical Contextual Analysis**
   - **Similarity Search:** Finds past changes with matching keywords or affected systems.
   - **Incident Correlation:** Adjusts risk upward if similar changes had incidents, or downward if historical rollouts were smooth.
   - **Data-Driven Reasoning:** Provides human-readable explanations like, "No prior incidents for this CI in the last 90 days — risk lowered."

3. **Automated Testing Mitigation Validation**
   - **CI/CD Integration:** Checks for existence and results of pre-production tests.
   - **Mitigation Checks:** Verifies backout plans or feature flags are in place when tests can't run.
   - **Dynamic Risk Adjustment:** Updates risk score and suggests next steps, e.g., "Proceed with caution; ensure rollback script is validated."

4. **Scheduling Conflict Detection**
   - **Calendar Scanning:** Detects overlapping changes on shared resources.
   - **Team Coordination:** Alerts relevant teams via chatops or email when conflicts arise.
   - **Recommendation:** Proposes rescheduling windows to minimize collision.

5. **Final Recommendation & Explainability**
   - **Risk Level:** Delivers a final green/yellow/red verdict.
   - **Actionable Guidance:** Provides a concise summary, e.g., "Yellow — deploy after verifying load-test results; rollback ready."
   - **Explainability Dashboard:** Visualizes contributing factors, historical data points, and step-by-step reasoning for stakeholder review.

Key Features & Benefits

- **Agentic Autonomy:** Minimal manual tuning—our AI agent learns and improves as more change outcomes are ingested.
- **Full Transparency:** Every recommendation is backed by a clear explanation, boosting trust and adoption.
- **Reduced Outages:** By catching high-risk changes early, we expect a significant drop in post-deployment incidents.
- **Scalable Extensible:** Modular design allows plugging in new data sources (e.g., performance metrics, error budgets).

Technical Stack

- **AI Agent Framework:** ServiceNow AI Agent Studio (or similar orchestration tooling)
- **Natural Language & ML:** Transformer models for text understanding, historical risk models for statistical insights

- **Data Fabric & Knowledge Graph:** Real-time access to change records, incidents, and CIs
- **Integrations:** CI/CD pipelines, testing platforms, chatops (Slack/MS Teams), and monitoring systems
- **Frontend:** Lightweight web dashboard for explainability and audit trails

Future Directions

- **Real-Time Observability Feeds:** Ingest live metrics to refine risk scores during the deployment window.
- **Cross-Project Learning:** Share anonymized learnings across teams to improve global model accuracy.
- **Self-Healing Automations:** Extend agents to automatically roll back risky changes when certain thresholds are breached.

Prepared for the "Innovation in SRE & Observability using Agentic AI" Hackathon.